

DataQual v3.0 — Comprehensive Platform Specification

AN EXTENSIVE GUIDE TO ANNOTATOR PROFILES, CONSENSUS METRICS & MULTI-PROJECT MLOPS

1. Platform Executive Summary

DataQual v3.0 is an enterprise Machine Learning Operations (MLOps) quality intelligence platform. It allows project managers and ML engineers to monitor data labeling accuracy, evaluate inter-annotator consensus, audit demographic parity, and manage active learning queues across multi-member evaluator teams.

2. Data Annotator Profile Specification (10 Core Fields)

DataQual captures 10 detailed information fields for every data annotator to enable accurate tier ranking and operational routing:

Field Name	Description & Purpose	Example Value
1. Full Name	Annotator's legal or display name.	Dr. Aris Thorne
2. Email Address	Contact and notification email.	aris.thorne@dataqual.ai
3. Role	Operational responsibility (Lead, Reviewer, Annotator).	Lead Evaluator
4. Specialization	Domain expertise (Medical, Vision, NLP, Fraud, 3D).	Medical Imaging (CV)
5. Seniority Level	Experience tier (Junior, Mid-Level, Senior, Lead).	Senior Expert
6. Native Language	Language proficiency for NLP / localization tasks.	English / German
7. Accuracy Score (%)	Benchmark accuracy against gold standard items.	98.2%
8. Throughput Target	Daily annotation volume target.	420 items/day
9. Hourly Rate (\$)	Operational cost allocation rate.	\$45.00 / hr
10. Status	Current availability (Active, On Break, Offline).	Active

3. Concurrent Multi-Project Support (5 Active Domains)

Project ID & Name	Domain / Category	Dataset Size	Target Accuracy
proj-1: Autonomous Driving Perception	Computer Vision	25,000 items	98.4%
proj-2: LLM Medical NLP Benchmark	NLP / Healthcare	18,500 items	96.2%
proj-3: E-Commerce Visual Search	Multimodal CV	42,000 items	94.8%
proj-4: Financial Fraud Risk Modeling	Tabular / Risk	12,000 items	99.1%
proj-5: Robotics Spatial Grasping	3D Point Cloud	30,000 items	92.5%

4. Feature Breakdown & Operational Workflows

- **Dashboard Overview:** Live KPIs (Total Annotators, Average Quality Score, Daily Throughput) and 7-day quality trend charts.
- **Ranking Leaderboard:** Sortable table ranking annotators into Platinum (≥ 90), Gold (75-89), Silver (60-74), and Bronze (< 60) tiers.
- **Consensus Engine:** Calculates Krippendorff's Alpha (α), Gwet's AC1, Dawid-Skene competence matrices, and manages the Adjudication Queue.
- **Smart Sampling:** 2D UMAP scatter projection of active learning acquisition queues (BADGE / BALD sampling algorithms).
- **AI Assist & Anchoring Safeguard:** Tracks human acceptance of LLM pre-annotations and triggers Blind Review mode if over-reliance $> 85\%$.
- **Fairness Audit:** Audits demographic parity, subgroup accuracy, and class label distribution skews.
- **Data Programming:** Programmatic weak supervision IDE for authoring Python Labeling Functions (Snorkel paradigm).
- **Data Input:** Complete 10-field annotator entry form + .CSV / .JSON file upload picker.
- **Settings & Refresh App:** Top-bar **Refresh / Reset App** button with safety confirmation modal.